

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q17-0820

New Salisbury 138 kV
July 24, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0820. The Interconnection Customer proposes a 637.1 MW natural gas combined cycle facility interconnecting at the New Salisbury 138 kV substation in the Virginia area. The request was received on July 02, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 360 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$132,394,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Frito Lay / Timberville 138 kV line	Blackberry / New Salisbury 138 kV	285.1	220	100.0
Timberville / New Salisbury 138 kV	Blackberry / New Salisbury 138 kV L	284.2	223	100.0
Blackberry / New Salisbury 138 kV L	Timberville / New Salisbury 138 kV	282.5	224	100.0
Chatom / Blackberry 138 kV	New Salisbury / Timberville 138 kV	276.9	227	100.0
Cedarville Junction / Detour 138 kV	St Louis Substation / Nichols 138 kV	211.7	248	20.2
Cedarville Jct / Page 138 kV	St Louis / Nichols 138 kV	210.3	252	20.2
Page / Purdy 138 kV L	Nichols / St Louis 138 kV	207.5	261	20.2
Tes Filer City Station Substation / Bridge River 138 kV Line	American / Urban 345 kV	204.5	206	11.4
Nuclear Fuels / Alvin 138 kV L	Sans Souci Substation / Oak Crk Station Substation 138 kV	193.8	201	15.7
Siloam Springs / Detour 138 kV	Detour / Ft Benton 138 kV	184.2	181	13.5
Nuclear Fuels / St Louis 138 kV	Roosevelt Rd / Wadsworth 138 kV Line	181.5	192	14.1
Crozet / Revital Substation 138 kV	Page / Purdy 138 kV	171.1	167	6.4
Bridge River / Abbotsford 138 kV L	Urban / American 345 kV Line	169.6	210	6.2
Cherry / Mount Jackson 138 kV	Chatom Substation / Purdy Substation 138 kV Line	166.9	178	11.6
Dpc Prairie Du Chien / Cherry 138 kV	Purdy Substation / Chatom 138 kV	159.6	192	11.6
Ft Benton Substation / Detour 138 kV	Siloam Sprngs / Detour 138 kV	158.3	250	14.6

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Roosevelt Rd / Elkton D.P 138 kV	St Louis / Nuclear Fuels 138 kV line	156.2	197	11.6
Alvin / Chatom 138 kV	Oak Crk Sta / Sans Souci 138 kV	154.7	208	15.7
Elkton D.P / Houston 138 kV	St Louis Substation / Nuclear Fuels Substation 138 kV	154.6	199	11.6
Roosevelt Rd / Wadsworth 138 kV	St Louis / Nuclear Fuels 138 kV	152.0	208	11.6
Edinburg / Springfield Power Llc 138 kV	Page / Purdy 138 kV	151.2	175	15.8
Revital / Dale Mabry 138 kV	Page / Purdy 138 kV	150.8	193	6.4
Dale Mabry / Gottoes 138 kV line	Purdy / Page 138 kV line	142.9	215	6.4
Lorin Industries Substation / Tampa Bay 138 kV	Tampa Bay / Dpc Prairie Du Chien 138 kV Line	141.2	181	18.6
Brooklyn City N Plant / Midet 138 kV	Abbotsford Substation / Bridge Riv 138 kV	140.7	227	8.7

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Frito Lay / Timberville 138 kV line	Rebuild line with higher-capacity conductor	\$5,171,000	100% (\$5,171,000)
Timberville / New Salisbury 138 kV	Rebuild line with higher-capacity conductor	\$5,204,000	100% (\$5,204,000)
New Salisbury / Blackberry 138 kV	Rebuild line with higher-capacity conductor	\$7,216,000	100% (\$7,216,000)
Blackberry / Chatom 138 kV	Rebuild line with higher-capacity conductor	\$8,641,000	100% (\$8,641,000)
Detour / Cedarville Jct 138 kV	Rebuild line with higher-capacity conductor	\$8,111,000	83% (\$6,761,000)
Cedarville Junction / Page Substation 138 kV Line	Rebuild line with higher-capacity conductor	\$10,934,000	83% (\$9,114,000)
Page / Purdy 138 kV	Rebuild line with higher-capacity conductor	\$2,384,000	83% (\$1,988,000)
Tes Filer City Station Substation / Bridge River 138 kV Line	Rebuild line with higher-capacity conductor	\$5,235,000	83% (\$4,331,000)
Nuclear Fuels / Alvin 138 kV	Rebuild line with higher-capacity conductor	\$7,076,000	92% (\$6,474,000)
Siloam Springs / Detour 138 kV L	Rebuild line with higher-capacity conductor	\$15,093,000	100% (\$15,093,000)
Nuclear Fuels / St Louis 138 kV	Rebuild line with higher-capacity conductor	\$7,454,000	74% (\$5,520,000)
Crozet / Revital Substation 138 kV	Rebuild line with higher-capacity conductor	\$5,239,000	48% (\$2,516,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Bridge River / Abbotsford 138 kV L	Rebuild line with higher-capacity conductor	\$11,641,000	67% (\$7,798,000)
Chery / Mount Jackson 138 kV	Rebuild line with higher-capacity conductor	\$4,714,000	100% (\$4,714,000)
Dpc Prairie Du Chien / Cherry 138 kV	Rebuild line with higher-capacity conductor	\$4,771,000	100% (\$4,771,000)
Ft Benton Substation / Detour 138 kV	Rebuild line with higher-capacity conductor	\$5,318,000	93% (\$4,959,000)
Roosevelt Rd / Elkton D.P 138 kV	Rebuild line with higher-capacity conductor	\$838,000	81% (\$678,000)
Alvin / Chatom 138 kV	Rebuild line with higher-capacity conductor	\$3,645,000	92% (\$3,334,000)
Houston / Elkton D.P 138 kV Line	Rebuild line with higher-capacity conductor	\$2,600,000	81% (\$2,105,000)
Wadsworth / Roosevelt Rd 138 kV	Rebuild line with higher-capacity conductor	\$4,264,000	81% (\$3,451,000)
Springfield Power Llc / Edinburg 138 kV	Rebuild line with higher-capacity conductor	\$4,209,000	94% (\$3,948,000)
Revital / Dale Mabry 138 kV	Rebuild line with higher-capacity conductor	\$8,412,000	48% (\$4,040,000)
Dale Mabry / Grottoes Substation 138 kV	Rebuild line with higher-capacity conductor	\$3,720,000	48% (\$1,786,000)
Lorin Industries Substation / Tampa Bay 138 kV	Rebuild line with higher-capacity conductor	\$4,358,000	94% (\$4,094,000)
Brooklyn City N Plant / Midet 138 kV	Rebuild line with higher-capacity conductor	\$10,512,000	83% (\$8,687,000)
	Total allocated to project		\$132,394,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.